CSCI 441 · DEMO 1

SuperAsk.AI

Sourced answers about Cambodian public services, in Khmer and English

PRESENTED BY

Yean · Prophea · Monunhh · Bunkheang · Heejae · Doungkhe

AGENDA

Who presents what

#	Presenter	Topic
1	**Yean**	The problem, our scope and Phase 0 status
2	**Prophea**	Using SuperAsk on the web and Telegram, live demo
3	**Monunhh**	Architecture and the answer safety pipeline
4	**Bunkheang**	Unit testing
5	**Heejae**	Integration testing and evaluation
6	**Doungkhe**	Data collection and the knowledge base
7	**Yean**	What’s next, then questions

THE PROBLEM

Finding what a procedure requires shouldn’t mean knowing someone

Scattered

Official information is spread across ministry sites, Facebook pages and Telegram channels.

Unsearchable

Much Khmer guidance is posted as images, often undated and without an owner.

Paid for

People lose a day with the wrong documents, or pay a broker for free information.

SuperAsk: ask in your own words, get the answer with its official source attached, or an honest “we have no verified source.”

PHASE 0 STATUS

A working prototype on web and Telegram

Working now

- Landing page and web chat, English and Khmer
- Ask API with retrieval, citations and guardrails
- Telegram bot and Telegram Mini App
- Feedback, honest declines and audit logs
- Deployed on Vercel, with CI on every push

In progress

- Real login (today it is a UI stub)
- Stronger retrieval models
- A larger, steward-approved knowledge base
- Follow-up memory in Telegram chats
- Content portal for ministry stewards

USING SUPERASK

One answer engine, three ways in

Web

Landing page plus the chat at /chat. Switch between ខ្មែរ and English at any time.

Telegram bot

Type a question in Telegram and get the same sourced answer as a chat reply.

Mini App

Open the full SuperAsk chat inside Telegram from the bot’s menu button.

Free to use, no account needed. Answers follow the language of the question.

LIVE DEMO

Five things we'll show you

1. **Ask on the web:** tap "Driving licence renewal" on the landing page
2. **Read the answer:** ministry header, steps, and every source with its review status
3. **Ask in Khmer:** the reply comes back in Khmer
4. **Ask something off-topic:** it politely declines instead of guessing
5. **Switch to Telegram:** open the Mini App from @superask168_bot

ARCHITECTURE

Every question passes eight gates, in order

1 · Before model

Guardrails

Refusal policy runs before any search

2 · Before model

Curated

Approved answers served word for word

3 · Before model

Retrieval

BM25 + dense search, fused, reranked

4 · Before model

Confidence

Weak matches decline, never guess

5 · Model

Generate

Writes only from the retrieved sources

6 · After model

Verify

Withholds any unsupported claim

7 · After model

Moderate

Checks the finished answer

8 · After model

Cite

Ministry, document, dates, freshness

TECH STACK

Swap the AI model with one setting

Built with

- Next.js 15 and React 19, TypeScript
- Tailwind CSS 4
- Telegram Bot API webhook
- Hosted on Vercel
- Only the server talks to a model; no key reaches the browser

Model provider	Runs where
SEA-LION (in use)	AI Singapore API
AI gateway	Third-party service
Claude API	External, outside Cambodia
Qwen via vLLM	Self-hosted in Cambodia
Extractive fallback	In-process, no model

All model code lives in lib/llm/. If a provider fails, answers fall back to extraction from the sources.

HONEST REFUSALS

When it won’t answer, the trace says why

Reason	What it means
policy	Not allowed: emergencies, legal advice, personal cases, politics
scope	Not a government-service question at all
retrieval	No approved source covers it, so it’s a content gap
verification	A written answer was withheld for an unsupported claim
moderation	The finished answer failed the output check

No hand-off: SuperAsk says “No approved source covers this” and stops.

UNIT TESTING

Every function checked on every push

610

unit tests, all passing

89.6%

line coverage; CI fails below 85%

0

paid model calls: CI runs with no AI credential

24 test files written with Vitest. Run locally with **npm test**; GitHub Actions runs them on every push and pull request.

UNIT TESTING · TEST MAP

Every stage of the answer pipeline has its own tests

Screen
141
guardrails 89
topicality 21
glossary 16
curated 15

→

Retrieve
148
query 33
fusion 30
normalize 26
content 25
romanize 23
stem 11

→

Decide
86
tiers 53
rationale 33

→

Generate
96
prompt 31
provider 27
gateway 22
sealion 16

→

Verify
21
verify 21

→

Cite + log
74
corpus 32
audit 21
freshness 13
trace 8

Plus 44 tests outside the pipeline: chat history 26, crawler 18. Total: **610 unit tests**.

UNIT TESTING · ONE TEST UP CLOSE

Catching an invented fee before a citizen sees it

```
// tests/engine/verify.test.ts
it("catches an invented fee", () => {
  const result = verify(
    "The fee is 45,000 riel.",
    ["X-1"],
    [source("X-1", "A fee of 30,000 riel
        applies on renewal.")],
    ["X-1"],
  );
  expect(result.passed).toBe(false);
  expect(result.unsupported).toContain("45000");
});
```

ARRANGE

The answer says **45,000** riel; the source says **30,000**

ACT

Run the verification gate on the answer

ASSERT

It fails, and names 45000 as unsupported

UNIT TESTING · COVERAGE BY FOLDER

Eight of nine folders clear the 85% floor



Orange line = 85%

CI fails the build below it. Overall: 89.6% lines, 94.1% branches, 96.0% functions.

Why lib/llm is lower

The Claude and vLLM adapters aren't in use yet. We test them when one becomes the provider.

UNIT TESTING · FROM PUSH TO MERGE

Nothing reaches main without passing CI

1 · Write code + tests
On a feature branch

2 · npm test
All 643 pass locally

3 · git push + PR
Pull request to main

4 · GitHub Actions
Runs CI automatically

Pass: merge to main

Fail: blocked, fix and push

INSIDE STEP 4 · TWO JOBS RUN IN PARALLEL

`check`	Install	Typecheck	**Tests, coverage ≥ 85%**	Build	Eval	4 gate checks
`secrets-scan`	No API keys committed	No .env file tracked				

PR #1, 4 Nov: secrets-scan passed in 4 s, check in 66 s, then merged

6 Oct: CI went red on the rebrand commit; fixed and green 6 min later

INTEGRATION TESTING

33 tests drive the real API routes end to end

Routes tested

- **POST /api/ask**: full pipeline, follow-up questions, audit record
- Provider failure degrades to a plain decline, never an error
- **POST /api/feedback**
- **GET /api/health**

CI gate, every push

1. Typecheck
2. All tests with coverage
3. Production build
4. Retrieval evaluation
5. Corpus and scope checks
6. Secrets scan

EVALUATION · 44-QUESTION GOLDEN SET

Both targets met

94.1%

Right source found in the top 8 (target 85%, passed)

100%

Off-limits questions refused (target 98%, passed)

67.6%

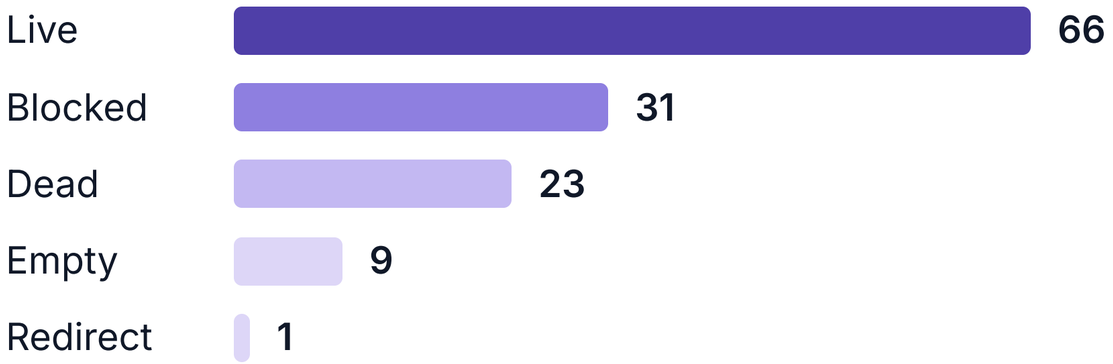
Right source ranked first, our next thing to improve

34 in-scope and 10 out-of-scope questions. Run with **npm run eval**; it runs in CI on every push.

DATA COLLECTION

We monitor 132 official government websites

LAST PROBE OF 130 SITES



Watch, never publish

The crawler flags changes for a ministry steward to review. Nothing it fetches becomes an answer on its own.

Ministries, regulators, 25 provinces, universities, hospitals and the two service portals.

THE KNOWLEDGE BASE

What SuperAsk can answer from today

45
source passages

5
curated answers

14
official terms explained

37
search aliases

Institution	Topics covered
Public Works and Transport	Driving licence, vehicle registration
Interior	Civil registration (birth certificates)
Commerce · National Bank	Trademarks · consumer finance

WHAT'S NEXT

From prototype to a service people rely on

Better ranking

Put the right source first more than 67.6% of the time

More approved content

A content portal so ministry stewards can approve sources

Telegram memory

Remember follow-up questions in Telegram chats

Real accounts

Replace the prototype login with real authentication

THANK YOU

Questions?

Yean · Prophea · Monunhh · Bunkheang · Heejae · Doungkhe

Try it: superask.vercel.app · Telegram @superask168_bot